**SIMATS ENGINEERING**

**ASSESSMENT TOOL 1 : Concept Mapping**

**COURSE NAME:** Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

**COURSE OUTCOME COVERED**

**CO1:** *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 50**
Question Count: 4

**Code of Conduct**

I ________**G VYSHNAVI(192525260)**____________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ________VYSHNAVI____________

**Assessment Tasks**

**Concept Mapping Tasks**

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

**Expected concept nodes:**

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer

- Infrastructure layer, platform layer, application layer, shared responsibility

**Concept Mapping Rubric**

| Criteria | Weightage | Level 4 - Exemplary | Level 3 - Proficient | Level 2 - Developing | Level 1 - Beginning |
|---|---|---|---|---|---|
| Concept Identification | 25% | All major concepts are accurately identified | Most concepts are identified correctly | Limited concepts are identified | Essential concepts are missing |
| Relationship Mapping | 25% | Links between concepts are accurate and meaningful | Most links are correct | Some links are weak or unclear | Relationships are mostly incorrect |
| Hierarchy and Organization | 20% | Excellent structure with clear hierarchy | Mostly organized with minor issues | Basic structure with weak hierarchy | No logical organization |
| Technical Relevance | 20% | Map strongly reflects cloud course content | Mostly aligned to topic | Partially aligned to topic | Weak alignment to topic |
| Visual Clarity | 10% | Neat, readable, and easy to interpret | Generally readable | Somewhat cluttered | Poorly presented |

**Permitted Tools :**

**draw.io / diagrams.net, Lucidchart, MindMeister, Miro, Canva, PowerPoint, Google Drawings, or equivalent diagramming tool**

Procedure for Preparing the Digital Concept Map

1. Use the following procedure to prepare the diagram and upload evidence.
2. Students who already know another diagramming tool may use it, but the exported evidence and source file must still be submitted.
3. Open https://app.diagrams.net/ or the desktop version of draw.io / [diagrams.net](https://diagrams.net).
4. Choose Device, Google Drive, or OneDrive as the save location according to availability.
5. Create a blank diagram and name the file as CSA15_CO1_AT1_RegisterNumber_Concept_Map.drawio.

6. Place the concept map title at the center or as a clearly visible anchor node.
7. Add major cloud nodes: cloud definition, cloud evolution, cloud architecture, service models, deployment models, virtualization, web services, SOAP, REST, APIs, elasticity, scalability, resource pooling, measured service, and emerging paradigms.
8. Connect nodes with labeled arrows such as uses, depends on, enables, scales through, communicates through, belongs to, supports, provides, consumes, or controls.
9. Use colors meaningfully: one color for service models, one for cloud characteristics, one for API/web-service concepts, and one for concept map title-specific elements.
10. Export the final diagram as PNG or PDF. Keep the editable .drawio source file also.
11. Paste the exported diagram image or screenshot in this DOCX under Task 1.
12. Upload the DOCX, exported PNG/PDF, and .drawio file to Google Classroom.

Task 1: Concept Map

Create a concept map in draw.io, Lucidchart, Miro, Figma, or any suitable diagramming tool. The map must connect cloud ecosystem, cloud actors, service models, deployment models, virtualization, web services, REST/SOAP APIs, scalability, elasticity, and the student's Project.

Task 2A: Service Model Responsibility Matrix

Prepare a technical matrix that compares SaaS, PaaS, IaaS, FaaS/serverless, database-as-a-service, storage-as-a-service, identity-as-a-service, and monitoring/logging services. For each selected service category, explain what is managed by the cloud provider, what is managed by the student application team, and why that split fits the Project.

Task 2B: Cloud Service Decision Matrix

Map Project modules to cloud services. Include authentication, frontend hosting, backend/API layer, database, object storage, analytics/logging, notification, AI/Python module, security, CI/CD, backup, and monitoring. Students should justify the most relevant service choices instead of forcing every possible cloud service into the design.

Task 3: API Flow, Elasticity, and Scalability Design

Draw a professional cloud-flow diagram for the Project modules. The diagram must show user app or web app, API gateway/backend service, authentication, compute layer, database, object storage, analytics/logging, monitoring, security boundaries, and scaling points. Students may replicate the given visual pattern, but all components and labels must be customized to their own architecture.

Task 4: Technical Justification

Write a concise justification explaining how the diagram and matrices prove CO1. The justification must discuss service model selection, API role, elasticity, scalability, virtualization or managed compute choice, and cloud responsibility boundaries.

| |
|---|
| 1. Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why? |
| 2. Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent? |
| 3. Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system? |
| 4. Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience? |
| 5. Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it? |

Expected Digital Evidence

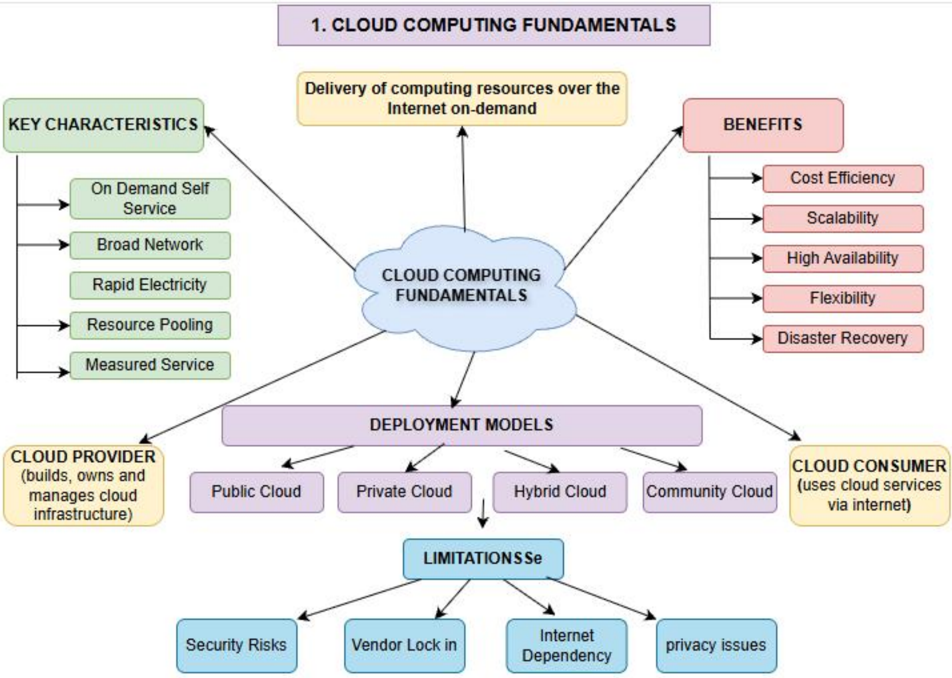
- Editable diagram source file and exported PNG/PDF for the concept map.
- Completed service model responsibility matrix and cloud service decision matrix.
- Architecture/API flow diagram exported as PNG/PDF.
- README file containing title, tools used, repository structure, and a short CO1 reflection.

GitHub and Google Classroom Submission

1. Create a repository using a clear name: CSA15_CO1_AT<Number>_<RegisterNumber>.
2. Upload editable source files, exported PDF/PNG evidence, screenshots, and a short README explaining the work.
3. Keep filenames professional and include the register number where appropriate.
4. Submit only the GitHub repository link in Google Classroom before the deadline.
5. Ensure the repository is accessible to the faculty evaluator during assessment.

Implementation Note

Faculty may evaluate the submitted evidence digitally using repository inspection, diagram quality, oral explanation, and CO-aligned rubric scoring. The focus is on technical understanding, cloud design reasoning, and hands-on evidence rather than routine written work.




**Concept Map 1: Cloud Computing Fundamentals**

**1. Effectiveness**

The concept map clearly presents the definition, characteristics, benefits, and limitations of cloud computing. The hierarchical design makes the concepts easy to understand. Grouping related ideas improves readability.

**2. Scalability, Elasticity & Resource Management**

Scalability and elasticity allow cloud resources to adjust based on demand. Resource management ensures efficient utilization of computing resources. Without them, performance, availability, and cost efficiency would decrease.

**3. APIs/Web Services/Cloud Services**

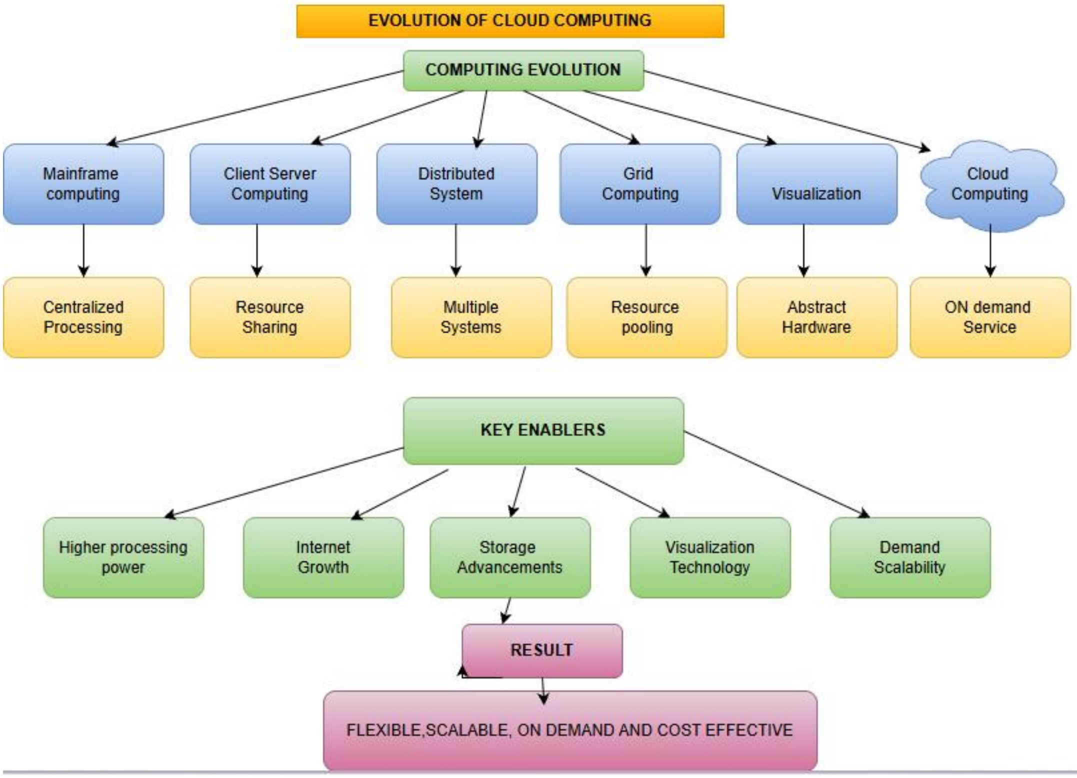
Cloud services such as SaaS, PaaS, and IaaS are included to show service delivery. APIs and web services enable communication between cloud applications. They improve flexibility and interoperability.

**4. Relationships Between Components**

The characteristics, benefits, and limitations are directly connected to the cloud computing concept. These relationships explain how cloud services operate efficiently. They improve user understanding and system reliability.

**5. Reflection**

Creating this map improved my understanding of cloud computing fundamentals. It helped visualize how different concepts are related. I would add deployment models and real-world examples in the future.




## Concept Map 2: Evolution of Cloud Computing

### 1. Effectiveness

The map clearly shows the progression from hardware evolution to cloud service delivery. Arrows illustrate the development of cloud technologies over time. This improves logical understanding.

### 2. Scalability, Elasticity & Resource Management

Modern cloud architecture evolved to support scalability and elasticity. Resource management became more efficient through distributed systems. Without these features, cloud computing would not support dynamic workloads.

### 3. APIs/Web Services/Cloud Services

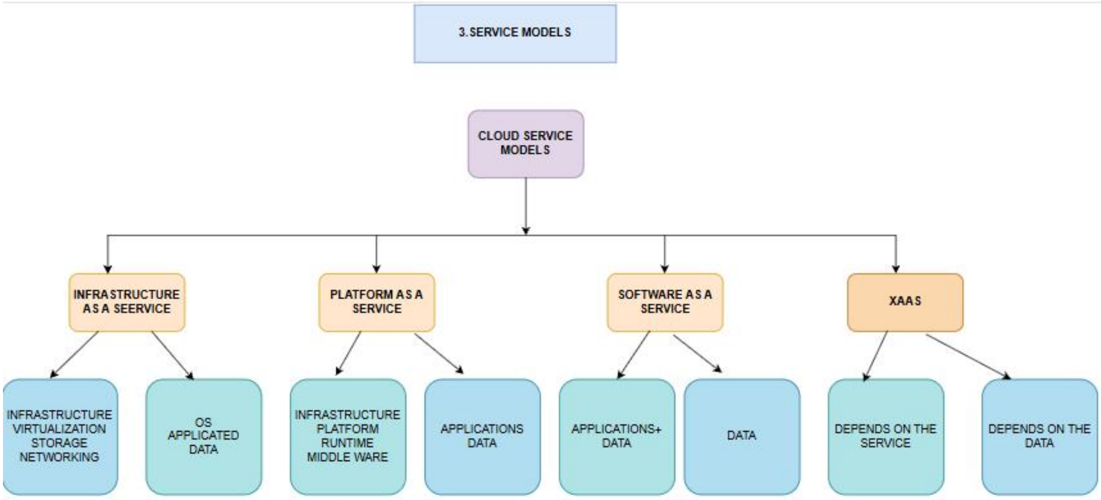
Internet software and web services enable communication between distributed systems. APIs connect applications across cloud platforms. These technologies make cloud services more efficient and accessible.

### 4. Relationships Between Components

Hardware evolution supports distributed systems, which enable cloud computing. Internet technologies connect these components to service delivery. Their interaction improves performance, reliability, and accessibility.

### 5. Reflection

Developing this map helped me understand how cloud computing evolved step by step. It connected historical technologies with modern cloud services. I would include timelines and major innovations for better visualization.




**Concept Map 3: Service Models**

**1. Effectiveness**

The map clearly differentiates IaaS, PaaS, SaaS, and XaaS. Provider and user responsibilities are shown separately. This makes the shared responsibility model easy to understand.

**2. Scalability, Elasticity & Resource Management**

Each service model uses scalable and elastic resources according to user needs. Resource management ensures efficient service allocation. Without these capabilities, cloud services would become slow and expensive.

**3. APIs/Web Services/Cloud Services**

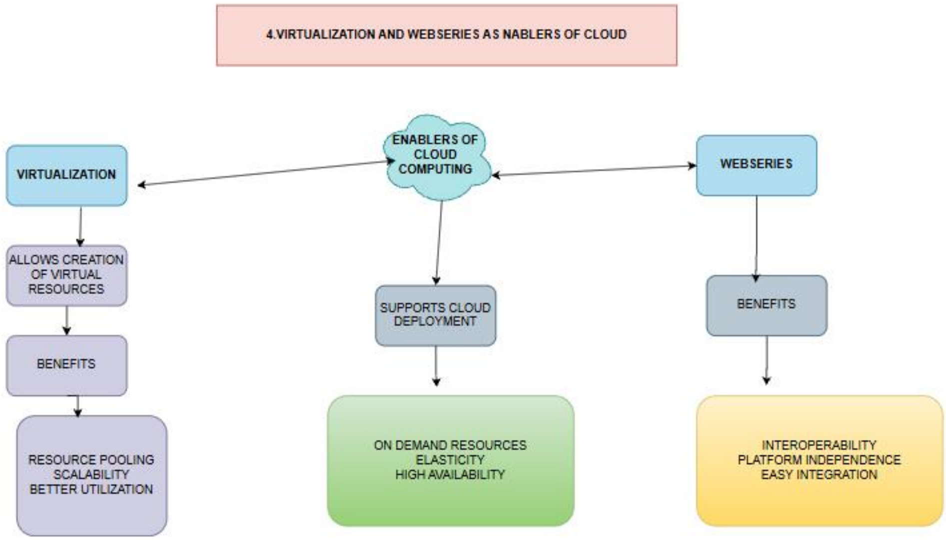
The service models represent different cloud service levels. APIs enable integration between users and cloud platforms. Web services support communication and application functionality.

**4. Relationships Between Components**

The provider manages infrastructure while users manage applications or data depending on the service model. These interactions improve security, flexibility, and reliability. They also simplify cloud service management.

**5. Reflection**

This concept map helped me understand the differences between cloud service models. It clarified the responsibilities of providers and users. I would add practical examples such as AWS EC2, Google App Engine, and Microsoft 365.




**Concept Map 4: Virtualization and Web Services as Enablers of Cloud**

**1. Effectiveness**

The concept map clearly shows virtualization and web services as the foundation of cloud computing. Their connection to cloud deployment and scalability is easy to understand. The layout effectively represents supporting technologies.

**2. Scalability, Elasticity & Resource Management**

Virtualization enables resource sharing, while elasticity adjusts resources automatically. Resource management optimizes computing performance and utilization. Without these features, cloud deployment would become inefficient and costly.

**3. APIs/Web Services/Cloud Services**

Web services and APIs enable communication between cloud applications. Virtualization provides the infrastructure required for cloud services. Together, they improve automation, scalability, and system efficiency.

**4. Relationships Between Components**

Virtualization provides virtual resources, while web services connect applications over the Internet. Their interaction supports reliable cloud deployment and better performance. This enhances user experience through faster and more flexible services.

**5. Reflection**

Creating this map helped me understand how virtualization and web services enable cloud computing. It highlighted their role in scalability and resource management. I would include containers and microservices to make the map more comprehensive.